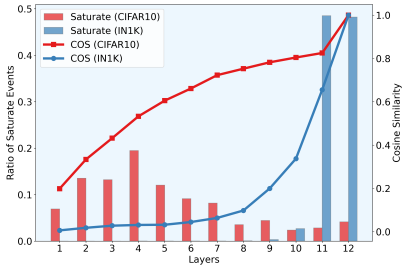


Figure 3: The DeiT-S models are trained on CIFAR10 and ImageNet-1K datasets from scratch. We measure the number of saturation events at each layer and their average cosine similarity with the final hidden states. More saturation events in early layers indicate higher cosine similarity.

Table 1: Comparison of the number of parameters across different architectures between multi-exit models with multiple classifiers (different classifiers at each layer) and a single classifier (ours). The last column shows that the percentage of saved parameters using single classifier.

Models	Hidden Dim	# of Classes	# of Layers	Multiple Classifiers (#Params)	Single Classifier (#Params)	#Param Saving
DeiT-S[24]	384	1,000	12	26.27M	22.05M	16.07%
DeiT-B[24]	768	1,000	12	95.02M	86.57M	8.89%
GPT-2[7]	768	50,257	12	541.57M	117.35M	78.39%
GPT-3[31]	12,288	50,257	96	233.67B	175.63B	25.10%
LLAMA-2 [56]	4,096	32,000	40	75.11B	70.35B	6.81%

3 Aligned Training for Enhancing Representational Similarity

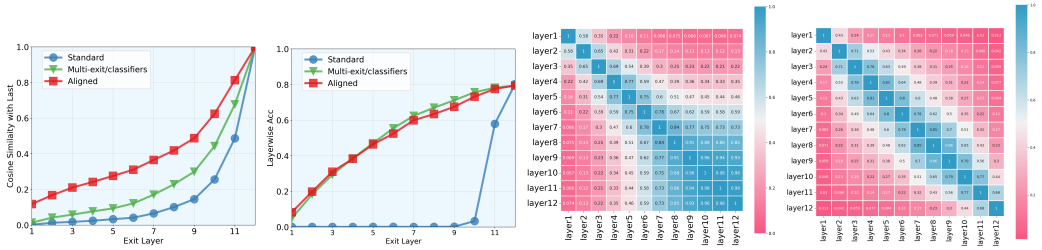
In the previous section, we observed that representations across layers within transformer models are positively correlated, resulting in saturation events when the last-layer classifier is directly applied after any hidden layer for early prediction and enabling a multi-exit model that shares a single classifier. In this section, we propose an aligned training method to enhance the effectiveness of shallow layers by improving layer-wise feature similarity. This, in turn, promotes more early saturation events, determines the minimal effective depth, and enhances performance when used as a multi-exit model. To the best of our knowledge, our work is the first to show that one common classifier is sufficient for multi-exit models. Table 1 shows a single classifier can significantly reduce the number of parameters and the computational complexity for multi-exit models, particularly for tasks with a large number of classes and large feature dimensions. Examples include ImageNet, with 1000 classes, and LLMs, where the number of classes equals the vocabulary size, i.e., the number of all possible tokens—for instance, the Llama-2 [56] model has a vocabulary of $32,000$ tokens while the GPT3 [31] has $50,257$ tokens.

3.1 Aligned Training for Enhancing Shallow Layer Performance

The ability to capture the layer-wise similarity of representations for each sample enables us to develop efficient methods for enhancing this similarity during training. A first approach is to directly add the cosine similarity between $\boldsymbol{h}^{(\ell)}$ and $\boldsymbol{h}^{(L)}$ for all $\ell < L$ as a regularization term during the training process. However, as shown in the appendix (see Figure 14), this approach can only slightly improve layer-wise similarity and accuracy. We conjecture this is due to the imbalance between the cross-entropy loss and the cosine similarity. Instead, motivated by the self-duality between the class-mean features and the linear classifiers, as observed in the $\mathcal{NC}$ phenomenon, we propose a simple yet efficient method, named aligned training, to enhance the layer-wise similarity by jointly optimizing the following aligned loss that is the weighted average of the CE loss from all the layers

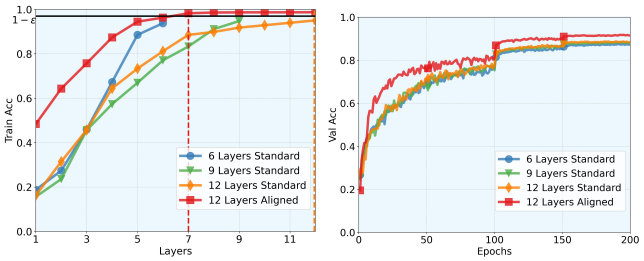
$$\mathcal{L}_{\text{aligned}}(\boldsymbol{x}, \boldsymbol{y}) = \sum_{\ell=1}^{L} \lambda_\ell \mathcal{L}_{\text{CE}}(\boldsymbol{W}\boldsymbol{h}^{(\ell)}, \boldsymbol{y}), \tag{5}$$

where $\boldsymbol{y}$ denotes the corresponding label for $\boldsymbol{x}$, $\lambda_\ell > 0$ is the weight for the ℓ-th layer. During the experiments, considering that shallow layers tend to have larger losses compared to deep layers,



(a) Cos Similarity (b) Layerwise Acc (c) Aligned (d) Multi-exit/classifiers

Figure 4: Comparison of ViT for ImageNet by standard training, proposed aligned training, and the multi-exit/classifiers, in terms of (a) cosine similarity, (b) layer-wise testing accuracy and (c-d) cosine similarities between all pairs of layers.



(a) Train Accuracy (b) Training Curve

Figure 5: Comparison of standard training of 6, 9, 12-layer DeiT-small model with aligned training of 12-layer model on CIFAR-10 in terms of (a) layer-wise train accuracy, and (b) convergence.

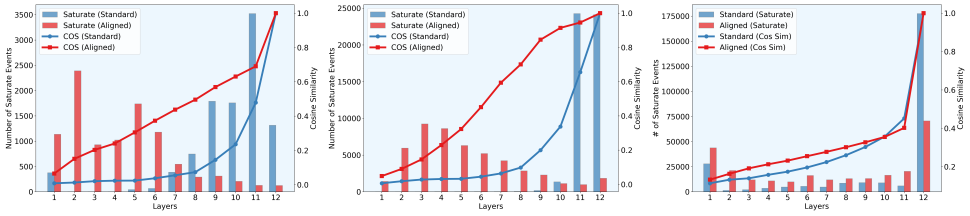
we set the weight to linearly increase with layers to put more emphasis on the deeper layers[5], i.e., $\lambda_\ell = 2\ell/(L(L+1))$. Roughly speaking, the aligned loss (5) introduces CE loss for intermediate layers and would encourage each layer features $\boldsymbol{h}^{(\ell)}$ to align with the common classifier $\boldsymbol{W}$—as implied by the $\mathcal{NC}$ phenomenon—hence improving the representation similarity across layers.

Figure 4 displays the layer-wise representation similarity and accuracy by the proposed aligned training. We can observe that **aligned training can significantly increase the layer-wise representation similarity and accuracy** by aligning all the features to the common classifier. Results in Appendix B.3 also show that aligned training can enhance progressive separation and compression from shallow to deep layers. To further illustrate the benefit of aligned training, we define the notion of ϵ-effective depth that modifies the notion exploited in Galanti et al. [58] by replacing nearest neighbor classifier accuracy with our layer-wise accuracy in (2).

Definition 3.1 *(ϵ-effective depth). We define the ϵ-effective depth $d^\epsilon(\mathcal{S}, f)$ of an L-layer transformer f over dataset $\mathcal{S}$ as the minimal layer ℓ such that $Acc_{\mathcal{S}}^{(\ell)} \geq 1-\epsilon$. Set $d^\epsilon(\mathcal{S}, f) = L$ if such ℓ is non-existent.*

Minimal ϵ-effective depth Denote the transformers learned by standard training and aligned training as f_{standard} and f_{aligned}, respectively. We observe that the aligned training yields a model with a much smaller ϵ-effective depth compared to standard training, i.e., $d^\epsilon(\mathcal{S}, f_{\text{aligned}}) < d^\epsilon(\mathcal{S}, f_{\text{standard}})$. Specifically, Figure 5(a) displays the layer-wise training accuracy of three transformers with different number of layers $\ell \in [6, 9, 12]$ with standard training and a 12-layer transformer with aligned training. For the standard training models, the train accuracy curves increase until the last layer without plateauing, even for the model with 12 layers, giving $d^\epsilon(\mathcal{S}_{\text{CIFAR10}}, f_{\text{standard}}) = 12$. While aligned training models unleash the power of shallow layers to transform features faster towards classifier across layers and push the redundancy behind, giving $d^\epsilon(\mathcal{S}_{\text{CIFAR10}}, f_{\text{aligned}}) \approx 7$, which is smaller

[5]Such a strategy of increasing weights is also employed in [57]. Uniformly weighting all layers (i.e., $\lambda_\ell = 1/L$) may diminish the importance of the deeper layers. We provide an ablation study for the comparison of linear increasing weights and uniform weights in Appendix B.3.



(a) DeiT on CIFAR10 (b) DeiT on ImageNet1K (c) GPT2 on WikiText103

Figure 6: Illustration of the effect of aligned training versus standard training on cosine similarity and the number of saturation events across vision (a,b) and NLP (c) models. Aligned training leads to more early saturation events by increasing the cosine similarity with the last hidden states.

than ϵ-effective depth of f_{standard}. On the other hand, the effective depth $d^{\epsilon}(\mathcal{S}, f_{\text{aligned}})$ increases with the complexity of the task, as demonstrated by comparing the results from CIFAR10 (Figure 5(a)) and ImageNet (Figure 4(b)). Thus, the effective depth, independent of the network's depth, can be leveraged to derive generalization bounds. This can be achieved by applying the approach from Galanti et al. [58], which offers non-trivial estimates of generalization based on effective depth.

Models f_{aligned} with small effective depth also offer several advantages in practical deployment. First, the layers beyond effective depth $d^{\epsilon}(\mathcal{S}_{\text{CIFAR10}}, f_{\text{aligned}})$ are redundant, as they do not contribute to accuracy improvements and can be pruned, leading to more efficient inference. Second, aligned training helps determine the minimal number of layers required for a task. While more complex tasks typically demand more layers, identifying the exact number can be challenging. In standard training, multiple models of different sizes must be trained to determine the minimal layer count. In contrast, with aligned models, retraining is unnecessary—one can simply apply the last-layer linear classifier to intermediate layers. Third, models trained using aligned method not only achieve slightly higher accuracy when truncated to 6 or 9 layers compared to models of the same size trained with standard methods (Figure 5(a)), but they also accelerate model convergence (Figure 5(b)) by providing immediate feedback to each layer, resulting in more effective parameter adjustments.

Saturation events and multi-exit model with a single classifier Another important application of the proposed aligned training is to improve the inference efficiency of large models. Figure 6 illustrates saturation events in both vision and NLP models (see the AlignedGPT setup in the appendix). We observe that aligned training **encourages earlier saturation events** by increasing the cosine similarity with the final hidden states. This demonstrates that f_{aligned} has greater potential for supporting early exits, commonly referred to as multi-exits in the literature [28–30]. In previous work, multi-exit models typically use separate classifiers for each layer, which can significantly increase the overall model size. As shown in Table 1, this issue becomes more pronounced when dealing with a large number of classes, as dense linear classifiers require a substantial number of parameters. To our knowledge, by exploiting the representation similarity within the transformer, *our proposed multi-exit model is the first to use a single classifier for all the layers*. In addition, when training multi-exit models, previous work [29] often uses additional KL-divergence terms to guide the logits of shallow layers by those of deep layers. In contrast, by using a common classifier to align shallow representations with deep ones, our aligned training does not require KL-divergence or other such terms. For comparison, we implement the multi-exit training with multiple classifiers [30], denoted by "multi-exit/classifiers", and display the results in Figure 4. On one hand, we observe that the proposed aligned training with a shared classifeir achieves higher layer-wise representation similarity than multi-exit/classifiers. On the other hand, the aligned training exhibits on-par performance as multi-exit/classifiers in terms of layer-wise accuracy, which is remarkable as the former only uses a single classifier.

To further show the performance of the proposed multi-exit models with a single classifier, we allow exit at shallow layers if the confidence level (max of softmax logits) exceeds a set threshold for each sample. On ImageNet dataset, Figure 7 displays the number of samples that exit at each layer. We observe that most samples exit at the last layer for standard training, while most exit at early layers for aligned training. We then calculate the classification accuracy along with the ratio of speed improvement measured by $\frac{\sum_{i=1}^{L} L \times m^i}{\sum_{i=1}^{L} i \times m^i}$ where m^i is the number of samples that exit at the i-th layer of DeiT. The model trained with standard training achieves 80.28% accuracy, while the model